

with a plastic sheet so that the PCB is completely isolated from 230V.

Results

The proposed system was tested on a refrigerator under real-life working conditions. The wall temperature profiles for fan-on and fan-off conditions were measured.

From the graphs shown in Fig.

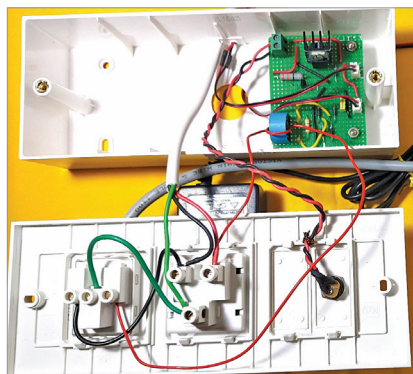


Fig. 9: Photograph showing the actual wiring done inside the switch box



Fig. 10: Photograph showing final system integration and fan attached to the fridge wall

11, it is observed that the peak temperature is almost 5°C lower when the fan is running. This reduction in fridge wall temperature provides following benefits:

1. Improved cooling efficiency, less food spoilage
2. Less load on the compressor, hence longer life
3. The user may be able to reduce the internal fridge temperature setting by one or two steps while maintaining the same cooling performance
4. Moderate power savings and a small reduction in the electricity bill

A small reduction in power consumption may save 25 to 50 rupees on the monthly bill. The actual savings will depend upon the following factors:

1. Refrigerator capacity in litres
2. Star rating (the lower the star rating, the higher the savings)
3. Free space around the fridge walls. If the gap is very small, forced cooling will provide better savings
4. Savings are likely to be higher in the summer
5. Heavy usage results in frequent fridge-door opening

Tips for further improvement

1. Instead of a fan with a plastic housing, select one with aluminium housing. The aluminium housing will absorb heat from the fridge

PARTS LIST

Semiconductors:

- T1 - BC546 NPN transistor
- T2 - TIP31C NPN power transistor (TO220 package)
- D1-D5 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 4.7 kilo-ohm
- R3 - 330-ohm, 1-watt

Capacitor:

- C1 - 0.47 μ F 100V box type

Miscellaneous:

- FAN1, FAN2 - 12V DC 0.25A, BLDC cooling fan 120mm $\times$ 120mm
- CT - Current transformer with ratio of 5A:5mA
- CON1, CON2 - Connector 2-pin polarised
- CON3 - 2-pin solderable terminals
- Heat sink - For TO220 package
- Magnets - Ring magnets 12mm OD, 4mm ID, 4mm thickness (4)
- Finger guards - For 120mm fans (2)
- Adaptor - 12V DC, 1A
- DC Jack - 12V DC socket for adaptor
- AC sockets - 3-pin 5A AC socket (2)
- Plastic film - A4 size plastic film (stiff)
- Switch box - For mounting 2 sockets

walls, which will then be removed by the fan airflow.

2. It is also observed that when the compressor and fans turn off, some heat remains stored in the walls. It takes a few more minutes to dissipate this stored heat. If the fan turn-off is delayed by a few minutes after the compressor has stopped, this heat can be dissipated faster.

3. For large-capacity refrigerators, four fans are recommended.

The proposed fan-cooling system is very easy to install. It improves the overall cooling performance of the refrigerator and offers moderate savings in the the electricity bill.

OEM refrigerator manufacturers can include this as an optional feature. In that case, the system cost will come down significantly, and even the CT will not be required.

The user can opt for this feature at the time of purchase by paying a small extra amount. For powering the fans, connectors could be provided on the back wall of the fridge. The user can then easily plug in or remove the fans as desired. **EFY**



Dr Vijay Deshpande currently works for solar energy projects. He has worked as electronics hardware engineer in several companies

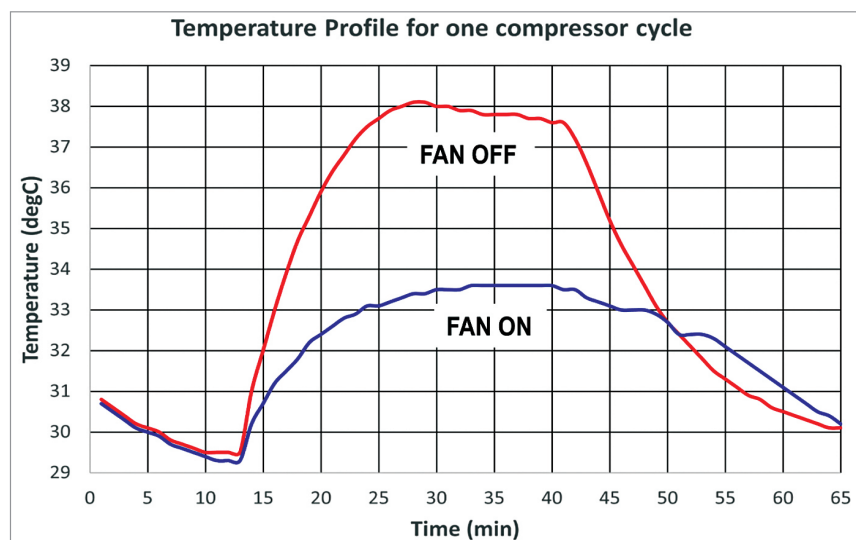


Fig. 11: Temperature profile of fridge surface; Red=Fan OFF, Blue=Fan ON